

Artificial Intelligence: A Short Overview

What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science focused on building systems that can perform tasks which normally require human intelligence. These tasks include understanding language, recognizing images, making decisions, and learning from experience. AI systems are trained on large amounts of data so they can identify patterns and make predictions or decisions with minimal human intervention.

Machine Learning

Machine Learning (ML) is a subset of AI that allows computers to learn from data instead of being explicitly programmed with fixed rules. Common types of machine learning include supervised learning, unsupervised learning, and reinforcement learning. Supervised learning uses labeled data to train models, for example predicting house prices from historical sales data. Unsupervised learning finds hidden patterns in unlabeled data, such as grouping customers into segments. Reinforcement learning trains an agent to make decisions by rewarding good actions and penalizing bad ones.

What is Retrieval-Augmented Generation (RAG)?

Retrieval-Augmented Generation, or RAG, is a technique that combines a language model with an external knowledge source, such as a collection of documents. Instead of relying only on what the model learned during training, a RAG system retrieves relevant pieces of text from a document database and passes them to the language model as context. This allows the model to answer questions using up-to-date or domain-specific information that it was never trained on, and reduces the chance of the model making up incorrect answers.

How RAG Works

A typical RAG pipeline has two phases. In the indexing phase, documents are loaded, split into smaller chunks, converted into numerical vectors called embeddings, and stored in a vector database. In the query phase, a user's question is also converted into an embedding, and the vector database is searched to find the most similar chunks. These retrieved chunks are then combined with the original question and sent to a large language model, which generates a final answer grounded in the retrieved content.

Why RAG is Useful

RAG is especially useful for building question-answering systems over private or specialized documents, such as company policies, research papers, or product manuals. It reduces hallucination because the model bases its answer on retrieved evidence rather than only its internal training data. It is also cheaper than retraining a model, since new information can simply be added to the vector database without changing the model itself.

Common Tools Used in RAG Systems

Popular tools for building RAG applications include LangChain and LlamaIndex for orchestration, ChromaDB and FAISS for vector storage, and embedding models from providers such as OpenAI, Hugging Face, or Cohere. Language models used for generation can be hosted models like GPT or Groq-hosted open models, or fully local models run through tools like Ollama.